

Dual Radar Imaging and Communications with Directional Sparse Transmit-Receive Beamforming

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Abstract—Dual radar imaging and communications (DRIC) system, which is capable of sensing the environment and delivering high-speed communication simultaneously, serves as a key enabling technology for beyond 5G and 6G. In this paper, we investigate the joint sparse multiple-input multiple-output (MIMO) array design and transmit-receive beamforming to balance high-resolution imaging and reliable downlink communications. A unified optimization framework is proposed to design directional transmit and receive beamformers for coherently synthesizing a high-fidelity point spread function, which is equivalent to image addition in the spatial domain rather than the traditional time domain. The communication quality-of-service (QoS) is ensured via a row-sparse constrained precoding matrix. To address the non-convexity introduced by the cardinality constraints, a sequential convex relaxation approach is developed. Numerical results demonstrate the effectiveness of the proposed DRIC system in terms of imaging resolution and communication QoS.

Index Terms—Dual radar imaging and communication, sparse MIMO array, directional beamforming, image addition.

I. INTRODUCTION

DUAL-FUNCTION platforms serve as a key enabler for beyond 5G and 6G (B5G/6G) networks, capable of simultaneous data transmission and environmental sensing [1], [2]. Among such platforms, dual radar imaging and communications (DRIC) systems merge these functionalities to achieve high hardware efficiency, supporting emerging applications like augmented reality [3].

By leveraging the high degrees of freedom (DoFs) inherent in beamforming, multiple-input multiple-output (MIMO) technology has emerged as a cornerstone of DRIC systems, enabling both enhanced data rates and superior imaging resolution [4], [5]. In pursuit of unprecedented data rates and imaging resolution, next-generation B5G/6G systems are expected to integrate extremely large-scale MIMO (XL-MIMO) architectures operating across wideband, high-frequency regimes [6]–[8]. However, the performance of conventional compact XL-MIMO arrays encounters severe physical restrictions, hardware overhead, and mutual coupling among antennas [9]–[11].

Sparse arrays (SAs) represent a promising structure for future MIMO-DRIC to reduce system overhead while main-

taining high resolution. By strategically optimizing antenna positions, SAs can synthesize virtual array apertures that significantly exceed their physical dimensions [6], [12]–[14]. Structured SAs which provide closed-form configurations calculated for specific tasks like direction-of-arrival (DOA) estimation, have been widely adopted in various applications [15]–[17]. However, recent advances in high-speed antenna switching have enabled the implementation of reconfigurable, unstructured sparse arrays which allow for the dynamic activation of specific antenna subsets, thereby offering the flexibility required to satisfy the disparate demands of dual-functional systems [18]–[21].

In this paper, we propose a directional sparse beamforming framework for the DRIC system. Specifically, the transmit beamforming matrix for imaging and the precoding matrix for communications are jointly optimized with shared row-sparsity to boost imaging performance while maintaining communication qualities-of-service (QoSs). The design of optimal receive beamformer is also considered to achieve a well-shaped imaging point spread function (PSF). As the formulated problem is non-convex due to the cardinality constraints, a sequential convex relaxation method is proposed to approximate the non-convex constraints and obtain the solution iteratively. Numerical results demonstrate the effectiveness of proposed DRIC system in concurrent high resolution imaging and reliable downlink communication.

The rest of this paper is organized as follows. The signal model of the DRIC framework is provided in section II. Section III discusses the problem formulation and solutions. Section IV presents the numerical results, and the conclusion is drawn in Section V.

II. SYSTEM AND SIGNAL MODEL

Consider a DRIC system, as illustrated in Fig. 1, which simultaneously serves K downlink CUs and illuminate a scene comprising Q scatterers. The dual-function platform employs a transmitter array of M_t antenna with an inter-element spacing of d_t and a receiver array of N_r antenna with an inter-element spacing of d_r . A shared-aperture design for dual-function enables hardware efficiency, where the transmitter serves both downlink communication and sensing, while the receiver array is dedicated to sensing in this paper. We first consider the

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system and system model with a fully populated uniform array configuration, which is later refined to explore a sparse MIMO transceiver design.

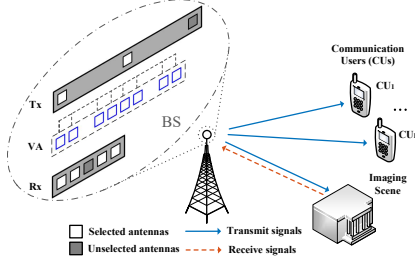


Fig. 1. Schematic diagram of the DRIC system.

Exploiting prior knowledge of the approximate scene direction, the array sequentially scans the environment by directionally transmitting and receiving multiple spatially focused beams, thereby maximizing energy concentration and mitigating spatial interference. For directional sensing targeted at the q -th scatterer, the joint transmit waveform is a superposition of the re-weighted radar waveform and the precoded downlink data stream, that is,

$$\mathbf{x}_q(t) = \mathbf{W}_{s,q}\mathbf{s}(t) + \mathbf{W}_c\mathbf{c}(t), \quad (1)$$

where $\mathbf{s}(t) \in \mathbb{C}^{I \times 1}$ is the orthogonal sensing waveform vector transmitted by the M_t antennas at the time instant t , $\mathbf{c}(t) \in \mathbb{C}^{K \times 1}$, $K < M_t$, is the downlink data stream. To ensure notation consistency, let $\mathbf{W}_c = [\mathbf{w}_{c1}, \dots, \mathbf{w}_{cK}] \in \mathbb{C}^{M_t \times K}$ denotes the communication precoding matrix, where $\mathbf{w}_{ck}$ is the weight vector for the k -th user. The transmit sensing beamforming matrix is denoted by $\mathbf{W}_{s,q} = [\mathbf{w}_{s1,q}, \dots, \mathbf{w}_{sI,q}] \in \mathbb{C}^{M_t \times I}$, and the corresponding receive beamforming matrix is defined as $\mathbf{W}_{r,q} = [\mathbf{w}_{r1,q}, \dots, \mathbf{w}_{rI,q}] \in \mathbb{C}^{N_r \times I}$.

A. Communication Model

The underlying scenario involves K single-antenna downlink CUs observing the output of a frequency flat Gaussian noise channel. The channel output at the K users at time instant t , represented via $\mathbf{y}_c \in \mathbb{C}^{K \times 1}$, is given by,

$$\mathbf{y}_{c,q}(t) = \mathbf{H}\mathbf{x}_q(t) + \mathbf{n}_c(t) = \mathbf{F}_c\mathbf{c}(t) + \mathbf{F}_{s,q}\mathbf{s}(t) + \mathbf{n}_c(t), \quad (2)$$

where $\mathbf{n}_c(t)$ is an additive white Gaussian noise (AWGN) vector with covariance $\sigma_c^2 \mathbf{I}_K$, $\mathbf{H} \in \mathbb{C}^{K \times M_t}$ is the downlink channel matrix, $\mathbf{F}_c = \mathbf{H}\mathbf{W}_c$ and $\mathbf{F}_{s,q} = \mathbf{H}\mathbf{W}_{s,q}$. In most cases, the channel state information (CSI) matrix can be estimated by transmitting the preamble sequence or is known to the CUs.

The SINR at the k -th CU, which directly relates to channel capacity [22], can be expressed by,

$$\gamma_{k,q} = \frac{|\mathbf{F}_c[k,k]|^2}{\sum_{j \neq k} |\mathbf{F}_c[k,j]|^2 + \sum_j |\mathbf{F}_{s,q}[k,j]|^2 + \sigma_c^2}, \quad (3)$$

where $[\bullet]_{k,j}$ denotes the entry of the matrix $\bullet$ located in the k -th row and the j -th column. In the sequel, the SINR of each downlink user is required to be no less than a pre-defined threshold Γ to guarantee a desired communication QoS, that is, $\gamma_{k,q} \geq \Gamma$, $k = 1, \dots, K$.

B. Imaging Model

The transmitted signals in Eq. (1) are reflected off the Q principal scatterers of the scene and observed by the receiver, which can be expressed as,

$$\mathbf{y}_{r,q}(t) = \mathbf{A}_r \text{diag}\{\boldsymbol{\eta}\} \mathbf{A}_t^T \mathbf{x}_q(t) + \mathbf{n}_r(t), \quad (4)$$

where $\boldsymbol{\eta} = [\eta_1, \dots, \eta_Q]^T$ is the vector consisting of the reflection coefficients of all Q scatterers from the direction θ_q , $\mathbf{n}_r$ is modeled as a zero-mean complex Gaussian white noise with a power of σ_r^2 , $\mathbf{A}_t = [\mathbf{a}_t(\theta_1), \dots, \mathbf{a}_t(\theta_Q)]$ and $\mathbf{A}_r = [\mathbf{a}_r(\theta_1), \dots, \mathbf{a}_r(\theta_Q)]$. Here, $\mathbf{a}_t(\theta_q)$ and $\mathbf{a}_r(\theta_q)$ are array steering vectors for transmitter and receiver, respectively.

As the radar receiver has complete knowledge of the transmitted communication symbols, the communication signal can be effectively removed and is thus not considered interference [23]. Performing matched-filtering to the received data vector in Eq. (4) yields,

$$\bar{\mathbf{Y}}_q = \int_T \mathbf{y}_{r,q}(t) \mathbf{s}^H(t) dt = \mathbf{A}_r \text{diag}\{\boldsymbol{\eta}\} \mathbf{A}_t^T \mathbf{W}_{s,q} + \mathbf{N}_r, \quad (5)$$

where T is the waveform duration, and $\mathbf{N}_r$ is the additive noise after matched filtering. In practical MIMO radar receivers, the signals observed at the output of the radar receiver are the $IN_r \times 1$ vector of virtual data, that is,

$$\bar{\mathbf{y}}_q = \text{vec}(\bar{\mathbf{Y}}_q) = \bar{\mathbf{A}}\boldsymbol{\eta} + \bar{\mathbf{n}}_r, \quad (6)$$

where the q -th column of $\bar{\mathbf{A}}$ is defined as $\bar{\mathbf{a}}(\theta_q) = \bar{\mathbf{a}}_t(\theta_q) \otimes \mathbf{a}_r(\theta_q) = [\mathbf{W}_s^T \mathbf{a}_t(\theta_q)] \otimes \mathbf{a}_r(\theta_q)$, $\otimes$ is the Kronecker product.

Applying the receive beamforming matrix $\mathbf{W}_{r,q}$ column-by-column to the matched-filtered virtual data vector in (6), the synthesized spatial signal corresponding to the q -th scatterer is obtained as,

$$\bar{y}_{r,q} = \sum_{i=1}^I \eta_q (\mathbf{w}_{si,q} \otimes \mathbf{w}_{ri,q})^H (\mathbf{a}_t(\theta_q) \otimes \mathbf{a}_r(\theta_q)), \quad (7)$$

which implies a superposition of I component images, mirroring the widely applied image addition strategy in the spatial domain rather than the traditional time domain [12], [17].

With $d_t = N_r d_r$, the process of forming each component image can be viewed as applying a pair of transmit-receive beamforming weights to a virtual array (VA), termed as sum co-array. With a fully populated uniform array configuration, the sum co-array consists of $M_t N_r$ elements with inter-element spacing being d_r , which can be described as a set of pairwise sums of transmit and receive element positions, $\mathcal{D}_\Sigma = \{d_{t,i} + d_{r,j} | d_{t,i} \in \mathcal{D}_t, d_{r,j} \in \mathcal{D}_r\}$, where $\mathcal{D}_t = \{d_{t,1}, \dots, d_{t,M_t}\}$ and $\mathcal{D}_r = \{d_{r,1}, \dots, d_{r,N_r}\}$ are two sets of transmit and receive antenna positions.

The transmit-receive PSF completely characterizes the system's resolution and interference suppression capabilities, thus serving as a key metric for a comprehensive assessment of its imaging performance. By incorporating the spatial image addition process, the PSF for forming a beam towards the q -th scatterer with I pairs of transmit-receive beamforming vectors is defined as,

$$\psi_q(\theta) = \sum_{i=1}^I (\mathbf{w}_{si,q} \otimes \mathbf{w}_{ri,q})^H (\mathbf{a}_t(\theta) \otimes \mathbf{a}_r(\theta)). \quad (8)$$

While Eq. (8) captures the spatial angular resolution, practical 2D imaging would employ temporal wideband waveforms (e.g., LFM signals) to attain range resolution. Pulse compression typically yields a range profile approximated by a sinc envelope, which can be decoupled from the underlying angular-domain PSF. The resulting 2D image is essentially the product of this range envelope and the angular PSF.

To sum up, by jointly designing the precoding matrix $\mathbf{W}_c$, the transmit beamforming matrices $\mathbf{W}_{s,q}$ and the receive beamforming vectors $\tilde{\mathbf{w}}_{r,q}$, $q = 1, \dots, Q$, a high-quality communication link can be created between the BS and CUs, while a well-designed PSF can be formulated to construct an accurate image of the scene.

III. PROBLEM FORMULATION

We introduce a joint transmit-receive sparse beamforming framework that concurrently maximizes communication throughput and radar imaging precision. We select M out of M_t transmit antennas and N out of N_r receive antennas, forming a sparse array at the transmit and the receive. Following the derivation in Section II-B, the sparse MIMO array can achieve sum co-arrays with MN virtual elements, implying a DoF of $\mathcal{O}(MN)$. This enables the dual-function system to achieve high spatial resolution.

When considering a sparse MIMO array configuration, it is noted that a common sparse array is required for multibeam transmitter where all beams share the same sparse configuration but have different sets of weights [24]–[26]. With shared transmit antennas for simultaneous downlink communication and sensing, the selected rows of $\mathbf{W}_c$ and $\mathbf{W}_{s,q}$ are required to exhibit the same sparse support. Regarding the receiver, the receive beamforming vectors $\tilde{\mathbf{w}}_{r,q}$ inherits a special group sparsity since an identical sparse receive array structure is employed for all transmit waveforms.

Considering the above properties and requirements, the sparse MIMO transceiver design can be formulated as,

$$\min_{\{\mathbf{W}_{s,q}, \mathbf{W}_{r,q}\}_{q=1}^Q, \mathbf{W}_c, \mathbf{z}_1, \mathbf{z}_2, \rho} \rho, \quad (9)$$

$$s.t. \quad \|\psi_{\text{opt},q} - \mathbf{A}^T \sum_{i=1}^I (\mathbf{w}_{si,q} \otimes \mathbf{w}_{ri,q})\|_2 \leq \rho, \quad (9a)$$

$$\gamma_{k,q} \geq \Gamma, k = 1, \dots, K, q = 1, \dots, Q, \quad (9b)$$

$$|\mathbf{w}_{si,q}| \preceq \mathbf{z}_1, |\mathbf{w}_{ri,q}| \preceq \mathbf{z}_2, \quad (9c)$$

$$\|\mathbf{z}_1\|_0 = M, \|\mathbf{z}_2\|_0 = N, \quad (9d)$$

$$\|\mathbf{w}_{c(m)}\|^2 \leq P_0, m = 1, \dots, M_t, \quad (9e)$$

$$\|\mathbf{W}_c\|_{2,0} = \|\mathbf{z}_1\|_0, \quad (9f)$$

where $\psi_{\text{opt},q}$ is a vector comprising the desired PSF samples, and matrix $\mathbf{A}$ is the sum co-array manifold. Constraints (9b) and (9e) guarantee the minimum SINR, Γ , and the per-antenna power limit, P_0 , respectively, where $\mathbf{w}_{c(m)}$ denotes the m -th row of $\mathbf{W}_c$. The boolean vectors $\mathbf{z}_1$, $\mathbf{z}_2$ and the L_0 -norm constraints in (9c)–(9f) jointly enforce that exactly M transmit

and N receive antennas are activated, while ensuring that the precoding matrix $\mathbf{W}_c$ and all I sensing beams rigorously share an identical sparse support. Specifically, $\|\mathbf{W}_c\|_{2,0} = \#\{m \mid \|\mathbf{w}_{c(m)}\| \neq 0, m = 1, \dots, M_t\}$ indicates the cardinality of the rows of $\mathbf{W}_c$.

Owing to the non-convexity of the constraints (9a)–(9d) and (9f), it is difficult to directly solve the sparse MIMO transceiver design problem. Nevertheless, the original problem can be simplified by utilizing a sequential convex relaxation to transform the problem into a convex form and solved iteratively. First, for the k -th user, by constraining $[\mathbf{F}_c]_{k,k} = \Gamma_k$, the communication QoS constraint (9b) can be rewritten as a quadratic form in terms of $\mathbf{W}_c$ as well as $\mathbf{W}_{s,q}$,

$$\mathbf{h}_{(k)}(\mathbf{W}_c \mathbf{W}_c^H + \mathbf{W}_{s,q} \mathbf{W}_{s,q}^H) \mathbf{h}_{(k)}^H + \sigma_c^2 \leq (1 + \frac{1}{\Gamma}) \Gamma_k^2, \quad (10)$$

where $\Gamma_k = \tau_0 \sqrt{(\sum_{k=1}^K \|\mathbf{h}_{(k)}\|^2)/M}$ represents the equivalent receive signal gain, $0 \leq \tau_0 \leq 1$ is a parameter to adjust between sensing and communication performance [27].

Regarding the cardinality constraints (9c), (9d) and (9f), a box constraint relaxation can be applied and the binary sparsity can be promoted utilizing a modified iterative reweighted L_1 -norm which consists of a set of judiciously designed reweights [26]. To tackle the non-convexity of the Kronecker product $\mathbf{w}_{si,q} \otimes \mathbf{w}_{ri,q}$ in Eq. (9a), we linearize it at the l -th iteration via its first-order Taylor expansion around the previous points $\mathbf{w}_{si,q}^{(l-1)}$ and $\mathbf{w}_{ri,q}^{(l-1)}$. Thus, the problem in the l -th iteration is reformulated as follows,

$$\min_{\{\mathbf{W}_{s,q}, \mathbf{W}_{r,q}\}_{q=1}^Q, \mathbf{W}_c, \mathbf{z}_1, \mathbf{z}_2, \rho} \rho + \mu_1 \mathbf{v}_1^{(l)T} \mathbf{z}_1 + \mu_2 \mathbf{v}_2^{(l)T} \mathbf{z}_2, \quad (11)$$

$$s.t. \quad \|\psi_{\text{opt},q} - \mathbf{A}^T \sum_{i=1}^I \tilde{\mathbf{w}}_{i,q}\|_2 \leq \rho, \quad (11a)$$

$$|\mathbf{w}_{si,q}| \preceq \mathbf{z}_1, |\mathbf{w}_{ri,q}| \preceq \mathbf{z}_2, \quad (11b)$$

$$i = 1, \dots, I, q = 1, \dots, Q, \quad (11c)$$

$$0 \preceq \mathbf{z}_1 \preceq 1, 0 \preceq \mathbf{z}_2 \preceq 1, \quad (11d)$$

$$\mathbf{1}^T \mathbf{z}_1 = M, \mathbf{1}^T \mathbf{z}_2 = N, \quad (11e)$$

$$\mathbf{h}_{(k)} \mathbf{w}_{c,k} = \Gamma_k, k = 1, \dots, K, \quad (11f)$$

$$\|\mathbf{w}_{c(m)}\|^2 \leq \kappa_0 P_0 [\mathbf{z}_1]_m, m = 1, \dots, M_t, \quad (11g)$$

where μ_1 and μ_2 are artificially set trade-off parameters between the imaging performance and the binary sparsity, $\mathbf{v}_1^{(l)}$ and $\mathbf{v}_2^{(l)}$ are reweighted coefficient vectors for the transmit and receive antennas from the l -th iteration, κ_0 is a scale parameter of the transmit precoding weights, $[\mathbf{z}_1]_m$ is the m -th element of $\mathbf{z}_1$. Due to space limitations, we omit the selection and updating of the above parameters, and direct readers to [27] for details.

Hence, the original non-convex problem is transformed into a convex form and converges to a local optima with reason-

able settings of the introduced parameters. The computational complexity of the sequential convex relaxation method is dominated by the number of optimization variables. With I waveforms, Q scatterers, and K CUs, the worst-case complexity per iteration is approximately $\mathcal{O}((IM_tN_r + KM_t)^{3.5})$, which is computationally tractable for convex solvers.

IV. NUMERICAL RESULTS

We consider a dual radar imaging and communication scenario utilizing a MIMO array comprising a 10-antenna ULA transmitter with an inter-element spacing of 7.5 wavelengths and a 15-antenna ULA receiver with an inter-element spacing of half wavelength. We select 7 antennas from the transmitter array and another 7 antennas from the receiver array to construct the sparse transceiver. The platform transmits I orthogonal waveforms to reconstruct a scene consisting of $Q = 43$ scatterers which is located at the broadside and also serves $K = 2$ CUs at 24° and 30° with path loss of -10dB. The PSF obtained with different values of I is synthesized as depicted in Fig. 2(a), compared with that of a ULA and an MRA with same number of antennas [28]. Note that for a fair baseline comparison, the communication precoders of the uniform and MRA arrays are also optimized by solving (11) under their fixed structural configurations. It is illustrated that, the proposed sparse MIMO transceiver achieves a significantly narrower mainlobe, underscoring the high resolution sensing. Moreover, via comparing the results for different values of I , it is evident that image addition contributes to sidelobe suppression, which leads to an overall better-designed PSF.

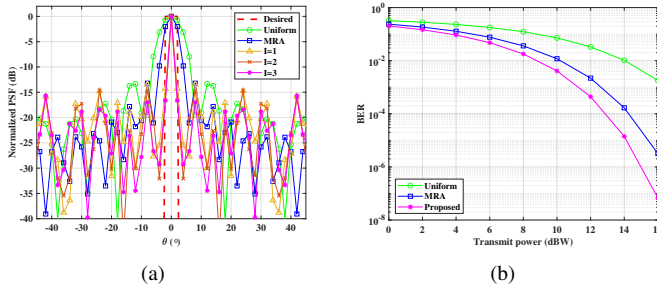


Fig. 2. (a) The normalized PSF for beam steering towards 0° and (b) BER versus communication power of different sparse MIMO transceiver configurations.

Next, we fix $I = 3$. The corresponding co-designed optimal sparse transceiver is shown in Fig. 3, where antennas on both ends of the transmit and receive arrays are selected to secure the largest virtual array aperture. The average BER of two CUs employing QPSK modulation is depicted for communication performance evaluation. We change the power of communication symbols, and obtain the corresponding achievable SINR with different transceivers, which is then used for conducting 10^8 Monte-Carlo trials to examine the average BER. The BER curves versus the communication power for different transceivers is depicted in Fig. 2(b). It is clear that the optimal sparse transceiver can achieve superior communication QoS benefiting from a high spatial resolution.

To provide a visual representation of the imaging performance of the proposed system, we conduct imaging on the scene depicted in Fig. 4(a) with the optimal sparse MIMO array and the MRA at SNR = 25dB. Canonical correlation and Pearson correlation coefficients are employed as metrics to assess the imaging performance [29]. The imaging results are shown in Fig. 4(b)-4(c) and the metrics are depicted in Fig. 4(d), demonstrating that the optimal sparse transceiver can reconstruct relatively higher-fidelity images.

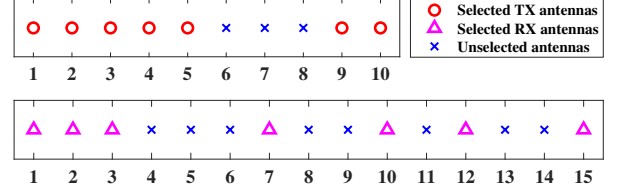


Fig. 3. Configurations of the selected transmit array (upper figure) and receive array (lower figure) corresponding to the optimal sparse MIMO array transceiver.

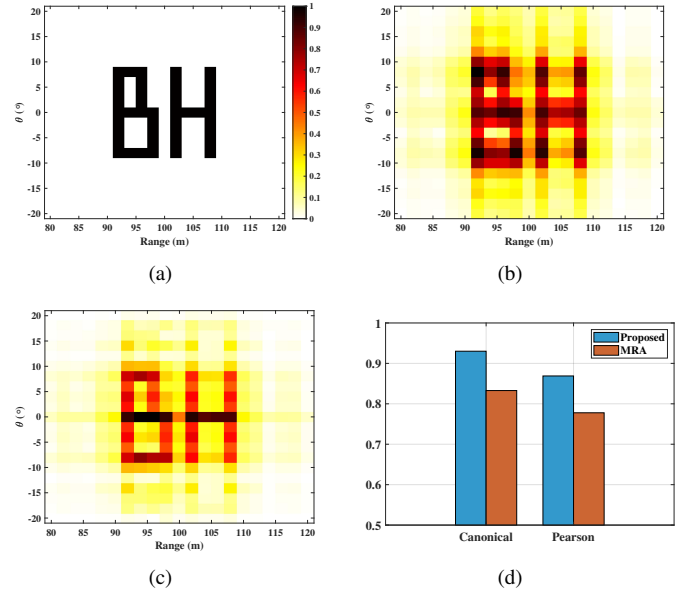


Fig. 4. (a) The scanned scene, (b) The image of MRA, (c) The image of the proposed sparse MIMO transceiver, (d) Canonical and Pearson correlation coefficients.

V. CONCLUSION

In this paper, we proposed a directional sparse beamforming design for dual radar imaging and communication systems. An optimization framework was formulated to synthesize a desirable point spread function based on directional transmit and receive weight pairs, while guaranteeing communication QoS via a group sparsity constraint. Furthermore, the non-convex constraints were addressed through a sequential convex relaxation approach. The effectiveness of the proposed method was demonstrated via numerical results.

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